2. Swift Fundamentals: Variables and Constants

Task 1:

Code:

```
import Foundation

let studentName = "Brinda"

let rollNumber: Int = 2301

let isHosteller = true

var semester: Int = 5

var cgpa: Double = 8.5

print("Name: \(studentName), Roll: \(rollNumber), Semester: \(semester)")

semester += 1

cgpa = 8.7

print("Name: \(studentName), Roll: \(rollNumber), Semester: \(semester), CGPA: \(cgpa)")

print(type(of: studentName))

print(type(of: rollNumber))

print(type(of: semester))

print(type(of: cgpa))

print(type(of: isHosteller))
```

Output:

Output

```
Name: Brinda, Roll: 2301, Semester: 5
Name: Brinda, Roll: 2301, Semester: 6, CGPA: 8.7
String
Int
Int
Double
Bool
```

Task 2:

Code:

```
import Foundation

var celsius: Double = 25.0

let fahrenheitFactor: Double = 9.0 / 5.0

let kelvinOffset: Double = 273.15

var fahrenheit = celsius * fahrenheitFactor + 32

var kelvin = celsius + kelvinOffset

print("Celsius: \(String(format: "%.1f", celsius))°C")

print("Fahrenheit: \(String(format: "%.1f", fahrenheit))°F")

print("Kelvin: \(String(format: "%.1f", kelvin)) K")

celsius = 37.5

fahrenheit = celsius * fahrenheitFactor + 32

kelvin = celsius + kelvinOffset

print("\nAfter updating Celsius:")

print("Celsius: \(String(format: "%.1f", celsius))°C")

print("Fahrenheit: \(String(format: "%.1f", fahrenheit))°F")

print("Kelvin: \(String(format: "%.1f", kelvin)) K")
```

Output:

Output

```
Celsius: 25.0°C
Fahrenheit: 77.0°F
Kelvin: 298.1 K

After updating Celsius:
Celsius: 37.5°C
Fahrenheit: 99.5°F
Kelvin: 310.6 K
```

Task 3:

Given Code:

```
let price = 250

var quantity: Int = 3

let taxRate = 0.18

price = 300

let total = price * quantity * taxRate

var title = "Order"

title = 42

print(total)
```

Output:

Output

```
ERROR!
/tmp/d1nux1sWOo-LMDqeyOD/main.swift:5:1: error: cannot assign to value:
'price' is a 'let' constant
price = 300
^~~~~
/tmp/d1nux1sWOo-LMDqeyOD/main.swift:1:1: note: change 'let' to 'var' to
make it mutable
let price = 250
^~~
var
ERROR!
/tmp/d1nux1sWOo-LMDqeyOD/main.swift:6:30: error: binary operator '*'
cannot be applied to operands of type 'Int' and 'Double'
let total = price * quantity * taxRate
```



Explanation:

1. Cannot assign to value: 'price' is a 'let' constant

- let price = 250 cannot be changed.
- Fix: Use var price = 250

2. Binary operator * cannot be applied to Int and Double

- price * quantity is an Int, but taxrate is Double. Swift cannot multiply them directly.
- Fix: Convert one value using Double(price) or declare price as Double.

3. Cannot assign value of type Int to type String

- title is a String, but 42 is an Int.
- Fix: Use title = “42”.

Correct Code:

```
import Foundation

var price: Double = 250.0

var quantity: Int = 3

let taxRate: Double = 0.18

price = 300.0

let subtotal = price * Double(quantity)

let tax = subtotal * taxRate
```

```
let total = subtotal + tax

var title = "Order"

title = "Updated Order"

print("Title: \(title)")

print("Price: \(String(format: "%.2f", price))")

print("Quantity: \(quantity)")

print("Subtotal: \(String(format: "%.2f", subtotal))")

print("Tax: \(String(format: "%.2f", tax))")

print("Total including tax: \(String(format: "%.2f", total))")
```

Output:

